Per Lundhammar

Clustering photon interactions in an event based detector

1 Introduction

1.1 Overview

PET and SPECT imaging relies on detecting photons and reconstructing their initial source position. We are considering an event-based detector where each incident photon produces a chain of interactions inside the silicon volume, each with timestamps, energy deposits, and 3D positions. However, at realistic radioactive source activities, many photons overlap in its measurements. The detector sees a stream of isolated interaction events, and the problem becomes: Which detector interactions belong to which photon?

For instance, consider two incident photons A and B in figure 1 that generates some number of interactions each in the detector volume.

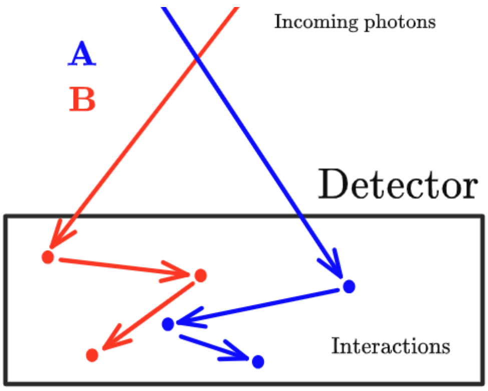

Figure 1: Two incident photons on the detector creating interacting in the detector creating measured times, positions, and lost energy at each interaction.

What we see is the measurement $d_i = (t_i, \mathbf{r}_i, E_i)$ of the measured interaction time t_i, 3D position $\mathbf{r}_i$, and the deposited energy E_i, see figure 2. So for a complete measurement series $D = [d_1, d_2, d_3, d_4, d_5, d_6]$ there is some true labeling $z_i \in (0, 1)$ such that we can associate each index $i \in (1, 2, 3, 4, 5, 6)$

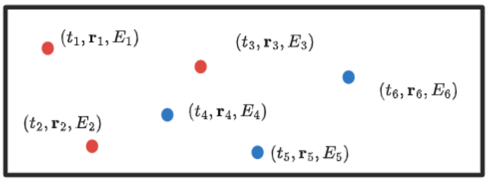

Figure 2: What is visible to us (except we don't know the true coloring of the interactions).

to one of these labeling. In this case the true labeling would be (letting red=0 and blue=1)

$$z_1 = 0, z_2 = 0, z_3 = 0, z_4 = 1, z_5 = 1, z_6 = 1.$$

1.2 Observed data

We simulated 15745 photons were detected, resulting in 39787 recorded interactions[1]. Each hit is stored as one row in a table. For any pair of interactions (i, j), we define the temporal, spatial, and energy distances

$$\Delta t_{ij} = |t_i - t_j|, \tag{1}$$
$$\Delta r_{ij} = \|\mathbf{r}_i - \mathbf{r}_j\|_2, \tag{2}$$
$$\Delta E_{ij} = |E_i - E_j| \tag{3}$$

For simulated data we also store a ground-truth photon identifier that links hits belonging to the same incident photon. Figure 3 illustrates the time ordering of hits in a short time window. Figure 4 compares pairwise time and spatial separations for hit pairs from the same photon versus different photons.

1.3 Bayesian distance-based clustering formulation

Note. Although deposited energy is recorded, it is not used for clustering in the present work.

Each interaction is assumed to originate from exactly one incident photon. We introduce a latent label

$$z_i \in \{1, 2, \dots\}$$

[1] Specifically, we simulated a point source at activity 500 MBq, embedded in the center of a water cylinder with radius 15 cm and length 70 cm in contact with a monolithic event based silicon detector with dimensions 20 cm × 20 cm × 5 cm with the larger face tangent to the cylindar surface.

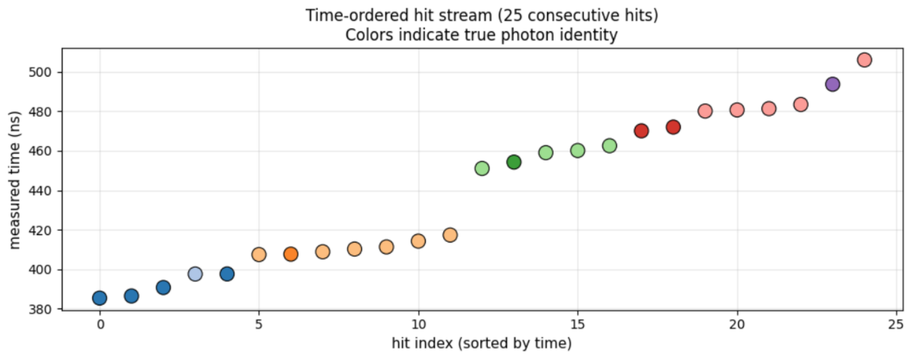

Figure 3: Hits in a short time window, sorted by time. In simulation, colors show the true photon identity.

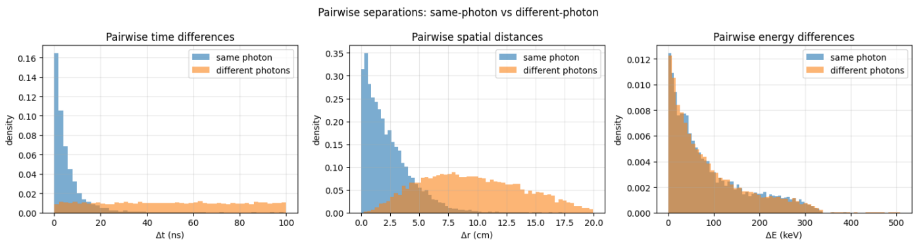

Figure 4: Pairwise separations Δt_{ij}, Δr_{ij}, and ΔE_{ij} for pairs from the same true photon versus different photons.

indicating which photon produced interaction i. The collection $z = (z_1, \ldots, z_N)$ defines a partition of the interactions into photon-induced clusters. The number of clusters is unknown. Our objective is to infer the posterior distribution over partitions z given the observed interaction data.

1.3.1 Within-cluster distance model

Following Dunson et al. [1], we define a likelihood only for distances between interactions assigned to the same cluster. Within each cluster, temporal and spatial distances are modeled using Gamma distributions with fixed shape parameter $\alpha_h = 2$ and cluster-specific scale parameters $\sigma_{t,h}$ and ${\sigma_{r,h}}^2$. Let

$$D^{[h]} = \{(\Delta t_{ij}, \Delta r_{ij}) : z_i = z_j = h,\ i < j\}$$

[2]Note: Dunson et al. place a Gamma prior on the shape parameter α_h to favor small modes while allowing moderate uncertainty. In the present work, we adopt a simplified formulation in which the shape parameter is fixed

denote the set of pairwise distances within cluster h. Conditional on cluster-specific scale parameters

$$\theta_h = (\sigma_{t,h}, \sigma_{r,h}),$$

we assume that within-cluster distances are independent and distributed according to

$$\Delta t_{ij} \sim g_t(\cdot \mid \sigma_{t,h}), \tag{4}$$

$$\Delta r_{ij} \sim g_r(\cdot \mid \sigma_{r,h}), \tag{5}$$

where g_t and g_r are Gamma densities with fixed shape parameter α_h and cluster-specific scale parameters. The contribution of cluster h to the likelihood is defined as

$$G_h(D^{[h]} \mid \theta_h) = \prod_{i<j: z_i = z_j = h} \left[g_t(\Delta t_{ij} \mid \sigma_{t,h}) \, g_r(\Delta r_{ij} \mid \sigma_{r,h}) \right]^{1/n_h},$$

where n_h is the number of interactions in cluster h. The exponent $1/n_h$ ensures that clusters of different sizes contribute on a comparable scale. The overall likelihood is then

$$L(D \mid z, \theta) = \prod_h G_h(D^{[h]} \mid \theta_h).$$

1.3.2 Prior on partitions

Partition prior. The number of photon-induced clusters in a measurement window is unknown and can vary from event to event. We therefore place a prior directly on the partition z rather than fixing the number of clusters in advance. Specifically, we use a Chinese Restaurant Process (CRP) prior with concentration parameter $\alpha_{\mathrm{CRP}} > 0$. This prior is exchangeable and assigns higher probability to partitions with fewer clusters, while still allowing new clusters to be created when supported by the data. The CRP thus provides a simple and flexible prior for modeling an unknown number of photons. In the present work, we use $\alpha_{\mathrm{CRP}} = 0.005$, which weakly favors fewer photon clusters while permitting additional clusters to form when interactions cannot be consistently explained by existing ones. This choice is appropriate for moderate activity levels, where interactions are sparse and over-segmentation is undesirable.

Distance model hyperparameters. For the distance distributions for Δt_{ij} and Δr_{ij}, the cluster-specific scale parameters are assigned independent inverse-gamma priors,

$$\sigma_{c,h} \sim \mathrm{InvGamma}(a_c, b_c), \qquad c \in \{t, r\}.$$

with $a_t = a_r = 2.0$ (the same as [1]), $b_t = 15.0$, and $b_r = 5.0$. These values correspond to prior means on the order of tens of nanoseconds in time and several centimeters in space, consistent with empirical separations observed for same-photon interaction pairs. Overall, these priors are intentionally weakly informative: they encode physically reasonable scales while allowing the posterior to be driven primarily by the observed interaction distances.

1.3.3 Posterior via Gibbs sampling

Posterior inference over the partition z and cluster-specific parameters $\theta = \{\theta_h\}$ is performed using Gibbs sampling. The sampler alternates between updating cluster assignments for individual interactions and updating cluster-specific scale parameters conditional on the current partition. After a burn-in period, samples from the Gibbs chain are used to estimate posterior quantities of interest. In particular, we compute the posterior co-assignment probabilities

$$P_{ij} = \mathbb{P}(z_i = z_j \mid D),$$

estimated as the fraction of retained samples in which interactions i and j are assigned to the same cluster.

1.3.4 Clustering accuracy metrics

Clustering performance is evaluated against simulation ground truth using two standard partition similarity measures: the Adjusted Rand Index (ARI) and the Variation of Information (VI).

ARI. This metric measures agreement between two partitions by comparing how pairs of interactions are grouped. It takes values in $[-1, 1]$, where ARI $= 1$ indicates perfect agreement and ARI $= 0$ corresponds to chance-level similarity. A negative ARI indicates less agreement than expected by chance [2].

VI. This metric equals zero if and only if the inferred clustering matches the ground truth exactly. In contrast to ARI, VI penalizes excessive fragmentation more strongly and is particularly sensitive to the creation of many small clusters. The VI metric scales as the $\log N$ where N is the number of clusters and we therefore report the normalized $VI/\log N$ score [3].

Together, ARI and VI provide complementary perspectives on clustering quality: ARI emphasizes pairwise correctness, while VI captures global structural agreement.

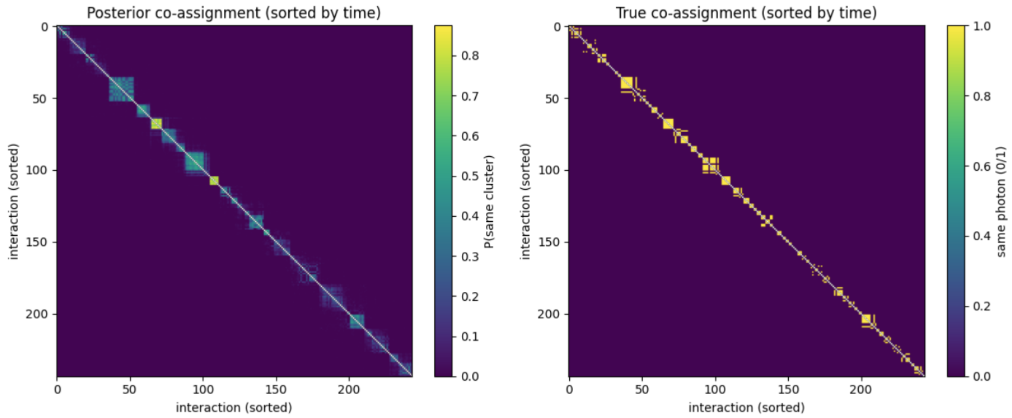

Figure 5: The co-assignment matrix (left) and the true cluster matrix (right).

2 Results and discussion

Due to computational reasons we restricted the data and considered only the first $N = 100$ photons of the simulation, which resulted in 244 interactions, with the goal of resolving the correct interaction cluster labels of these photons.

2.1 Posterior co-assignment structure

Figure 5 (left) shows the posterior co-assignment matrix $P_{ij} = \mathbb{P}(z_i = z_j \mid D)$ for a subset of interactions, sorted by measured time. Each entry represents the posterior probability that two interactions originate from the same photon.

The matrix exhibits a clear block structure, indicating groups of interactions that are consistently assigned together across posterior samples. Off-diagonal uncertainty reflects ambiguous interaction pairs, typically arising from temporal overlap between different photons. For comparison, Figure 5 (right) shows the corresponding ground-truth co-assignment matrix.

2.2 Thresholded clustering and clustering accuracy

To obtain a representative clustering for evaluation, we convert the posterior co-assignment matrix into a hard partition using probability thresholding. Specifically, we define an undirected graph in which interactions i and j are connected if

$$P_{ij} \geq \tau,$$

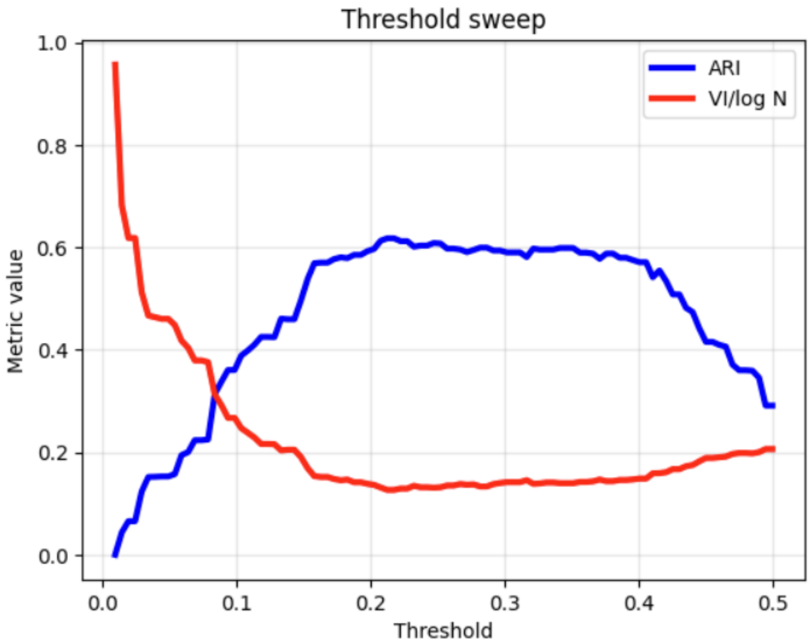

Figure 6: The clustering accuracy as measured by the ARI and VI for different thresholds τ.

where $\tau \in (0, 1)$ is a fixed probability threshold. The connected components of this graph define the inferred clusters. The inferred clustering is compared against simulation ground truth using the Adjusted Rand Index (ARI) and the Variation of Information (VI). In figure 6 the ARI and (normalized) VI is shown for different thresholds τ. We see that there exists a threshold for which the ARI ≈ 0.6 and the $\mathrm{VI}/\log N \approx 0.15$.

There is no absolute threshold that defines "good" clustering performance in terms of ARI, as its magnitude depends on the number of clusters, cluster sizes, and data overlap. Nevertheless, in applied clustering practice, ARI values between 0.0 and 0.5 are often taken to indicate moderate similarity, and values above approximately 0.5–0.7 are interpreted as indicating good similarity between inferred and true partitions [4]. Likewise, normalized VI values near zero indicate close structural agreement [5]. In our results, ARI ≈ 0.6 and $\mathrm{VI}/\log N \approx 0.15$ therefore reflect a good recovery of the underlying photon interaction structure.

3 Outlook

Several extensions of the present framework are possible. First, additional distance channels, such as energy differences or inferred scattering angles, could be incorporated as separate likelihood components. This would allow the model to adaptively weight different physical cues. Second, more structured priors could be introduced to reflect known properties of photon interaction chains, such as approximate ordering in time.

4 Conclusion

We have presented a simplified Bayesian distance-based clustering approach for grouping photon interaction chains in an event-based detector. Using posterior co-assignment probabilities inferred via Gibbs sampling, we obtained clusterings that recover the dominant interaction structure in simulated data, achieving an ARI of approximately 0.6 for 100 overlapping photons.

References

[1] L. L. Duan and D. B. Dunson, "Bayesian distance clustering," *Journal of Machine Learning Research*, 22(224):1–27, 2021. `http://jmlr.org/papers/v22/20-688.html`.

[2] Emergent Mind, "Adjusted Rand Index (ARI) explained," *Emergent Mind*, accessed January 2026. `https://www.emergentmind.com/topics/adjusted-rand-index-ari`.

[3] Wikipedia contributors, "Variation of information," *Wikipedia*, The Free Encyclopedia, accessed January 2026. `https://en.wikipedia.org/wiki/Variation_of_information`.

[4] A. Lun, "Clustering redux," in *Orchestrating Single-Cell Analysis with Bioconductor*, 2022. `https://bioconductor.org/books/3.14/OSCA.advanced/clustering-redux.html`.

[5] M. Meilă, "Comparing clusterings—an information based distance," *Journal of Multivariate Analysis*, 98(5):873–895, 2007.